



Policy as a Full Mediator Between Improvement in Construction Waste Management and C&DW Management Performance: PLS-SEM Evidence from Malang City, Indonesia

Hariyono Seputro Youngky Pratama^{1*}, Antariksa², Anthon Efani³, Wisnumurti⁴

¹ Doctoral Program in Environmental Science, Brawijaya University, 65115 Malang, Indonesia

² Department of Architecture, Faculty of Engineering, Universitas Brawijaya, 65145 Malang, Indonesia

³ Department of Socio-Economic Fisheries and Marine Science, Faculty of Fisheries and Marine Science, Universitas Brawijaya, 65145 Malang, Indonesia

⁴ Department of Civil Engineering, Faculty of Engineering, University of Brawijaya, 65145 Malang, Indonesia

* Correspondence: Hariyono Seputro Youngky Pratama (hariyonosyp@student.ub.ac.id)

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Abstract: This study examined the mediating role of policy in the relationship between improvement in construction waste management and effectiveness of construction and demolition waste (C&DW) management in Malang City, Indonesia. Using Partial Least Squares Structural Equation Modeling (PLS-SEM) with data from 48 construction companies, this research investigated how policy frameworks translated improvement initiatives into sustainable waste management outcomes. Improvement in construction waste management was measured by four phases (planning, design, construction, and operation), policy through four instruments (regulation, incentive, reward-sanction, and standard), and C&DW management through triple bottom line dimensions (economic, social, and environmental). Results revealed that improvement in construction waste management significantly influenced policy formulation ($\beta = 0.761$, $p < .001$, $f^2 = 1.374$), and that policy substantially affected the effectiveness of C&DW management ($\beta = 0.692$, $p < .001$, $f^2 = 0.950$). However, the direct effect of improvement in C&DW management was not significant ($\beta = 0.240$, $p = .249$), indicating full mediation through policy ($\beta = 0.526$, $p < .001$, $v = 0.077$). The model explained 78.8% variance in C&DW management with strong predictive relevance ($Q^2 = 0.711$). These findings demonstrated that improvement efforts should be channeled through robust policy frameworks to achieve systemic waste management transformation, thus highlighting the critical role of integrated policy instruments in translating operational improvements into sustainable outcomes in developing urban contexts.

Keywords: Construction waste management; Policy mediation; Circular economy; Partial Least Squares Structural Equation Modeling; Malang City

1. Introduction

The construction industry has emerged as a significant contributor to global environmental challenges, generating approximately 30–40% of total solid waste worldwide and contributing to 39% of global energy-related carbon dioxide emissions (Al-Otaibi et al., 2022; United Nations, 2021). Construction and demolition waste (C&DW) represents a critical environmental concern in both developed and developing nations, accounting for 36% of total waste generation in the European Union, 30% in the United States, 35% in Canada, and reaching 50% in the United Kingdom (Nawaz et al., 2023). In 2020, global C&DW production was dominated by China with 1,130 million tons, followed by 534 million tons in the United States, 86 million tons in Germany, and 85 million tons in the United Kingdom (Mohammed et al., 2021). This substantial waste generation not only places enormous pressure on landfill capacity but also results in significant environmental degradation and resource depletion, requiring urgent action toward sustainable waste management practices.

The transition toward principles of circular economy in construction has become imperative to mitigate

environmental impacts and reduce resource consumption. Research indicated that 50–80% of C&DW possessed potential for recycling and reuse when construction waste was properly sorted and managed, thus presenting substantial opportunities for the reduction of environmental impact (Kofoworola & Gheewala, 2009; Tanthanawiwat et al., 2024). The typical composition of C&DW includes concrete (12–40%), brick masonry (8–54%), asphalt (4–26%), wood (2–4%), metals (0.2–4%), gypsum (0.2–0.4%), plastics (0.1–2%), minerals (2–9%), and other materials (2–36%) (Gálvez-Martos et al., 2018). However, the environmental performance of C&DW recycling depends significantly on transportation to recycling facilities, separation processes, recycling technologies, and the quality of recycled materials (Waskow et al., 2021). These factors underscore the complexity of implementing effective C&DW management systems and the critical role of policy interventions in facilitating sustainable practices.

In Indonesia, particularly in rapidly urbanizing cities like Malang, C&DW management presents unique challenges that intersect with the broader municipal solid waste crisis. The construction sector in Malang has expanded significantly alongside urban growth, yet C&DW is not systematically separated from the general waste stream. Municipal data from 2023 show that fractions commonly associated with construction activities, i.e., wood/branches (13.6%), plastics (13.66%), metals (0.98%), and glass (1.78%), constitute approximately 30% of the City's solid waste in the National Waste Management Information System (SIPSN, 2025), hence suggesting a substantial but unquantified contribution from construction and demolition activities. This absence of differentiated C&DW tracking represents a critical gap in waste management data and policy formulation, impeding targeted interventions for construction-sector waste reduction and resource recovery.

Recent research has identified policy factors as significant mediating variables in the relationship between improvement efforts and sustainable construction waste minimization (Mohammed et al., 2022). However, policy factors do not necessarily mediate the relationship between current waste generation practices and waste management outcomes, thus suggesting that effective policies require support from improvement factors such as preventive planning and innovation rather than merely regulating existing practices. Studies have demonstrated that integrated policy approaches combining incentives, guidelines, and obligations prove more effective than single-policy instruments (Cheng et al., 2022; Wang et al., 2021). Experiences of implementation obtained from Malaysia and Hong Kong indicate that integrated policies could significantly reduce waste volumes directed to landfills (Mohammed et al., 2022; Wu et al., 2020). Economic instruments including taxes, subsidies, and disposal fees have also been shown to promote improved waste management behaviors (Cheng et al., 2022; Gálvez-Martos et al., 2018). Nevertheless, policy effectiveness remains heavily influenced by regulatory enforcement, stakeholder awareness and attitudes, and infrastructural support (Islam et al., 2024; Unegbu & Yawas, 2024; Zhao, 2021;).

This study aims to examine the mediating role of policy in the relationship between improvement in construction waste management and effectiveness of C&DW management in Malang City, hence providing evidence-based insights for policy formulation and implementation in similar developing urban contexts. Despite growing international recognition of policy's mediating role in C&DW management, limited empirical evidence exists regarding policy effectiveness in developing Indonesian cities. Studies in Malang highlighted the importance of collaboration of multiple stakeholders, community empowerment, and digitalization in waste management, while identifying key barriers including inadequate facilities, minimal community participation, and weak institutional coordination (Anisykurlillah et al., 2024; Gunawan, 2025). The transition from policy control models to co-creation approaches has been recommended (Gunawan, 2025), while digitalization and recognition of roles played by the informal sector have been emphasized as accelerators toward circular economy and SDG achievement (Kurniawan et al., 2022). Understanding of how policy factors mediate these relationships in Malang's specific socio-economic and institutional contexts remains limited; empirical investigation is thus considered to be essential.

2. Literature Review

Improvement of construction waste management represents a comprehensive and systematic approach to enhancing waste reduction, handling, reuse, recycling, and disposal practices throughout the lifecycles of construction projects. This improvement encompasses integrated processes of collection, transportation, storage, processing, recovery, and disposal designed to achieve acceptable environmental quality standards while supporting sustainable development objectives (Kurniawan et al., 2024; Ting et al., 2022). The operationalization of construction waste management improvement occurs through four critical and interconnected phases that collectively determine overall effectiveness. First, the planning phase establishes the foundation through strategies of preventive planning, target setting of waste reduction, and planning of material optimization that anticipate and minimize waste generation before the commencement of physical work (Ajayi et al., 2017; Kabirifar et al., 2020). Second, the design phase translates planning intentions into concrete specifications through design for waste minimization principles, material selection optimization that prioritizes recyclable and low-waste materials, and design modifications that reduce cutting waste and material surplus (Liu et al., 2020; Shajidha & Mortula, 2025). Third, the construction phase implements practical waste management through on-site waste sorting systems,

efficient construction practices that minimize material wastage, and real-time monitoring systems that track waste generation and enable rapid corrective actions (Zhuang et al., 2020). Fourth, the operation phase extends considerations of waste management beyond project completion through maintenance practices that prolong material lifespan and plan for end-of-life material recovery that facilitates recycling and reuse when buildings are eventually renovated or demolished (Schützenhofer et al., 2022). These four phases align fundamentally with the hierarchical 3R principles (Reduce, Reuse, and Recycle) and the framework of circular economy, which emphasize source reduction as the most environmentally and economically effective strategy while recognizing the complementary importance of reuse and recycling initiatives (Alhawamdeh et al., 2024; Kabirifar et al., 2020). Research demonstrated that integrated implementation across all four phases yielded superior outcomes compared with isolated interventions, as synergies between planning foresight, design optimization, construction discipline, and operational stewardship create compounding benefits in waste reduction, cost savings, and environmental protection (Kurniawan et al., 2024; Shajidha & Mortula, 2025).

Policy frameworks serve as essential regulatory and incentive architectures that govern, facilitate, and accelerate effective C&DW management through diverse instrumental mechanisms. These policy instruments operate through four primary and often complementary channels that shape stakeholder behaviors and organizational practices. First, reward and sanction systems establish structures of accountability through performance-based reward mechanisms that recognize and incentivize exemplary waste management practices while imposing penalty mechanisms for non-compliance, thereby creating obvious behavioral expectations and consequences that motivate sustained adherence to requirements of waste management (Ding et al., 2023; Wang et al., 2021). Second, incentive mechanisms reduce economic barriers to sustainable practices through financial instruments including tax reductions for waste recycling activities, direct subsidies for establishing or utilizing recycling facilities, and preferential treatment in the processes of public procurement for contractors demonstrating strong waste management performance, which collectively make sustainable practices economically attractive rather than burdensome (Cheng et al., 2022; Gálvez-Martos et al., 2018; Hua et al., 2022). Third, regulatory frameworks establish mandatory baseline requirements through legally binding provisions including preparation and submission of mandatory waste management plan, obligatory on-site waste sorting and separation practices, and landfill disposal restrictions that progressively reduce acceptable waste streams, thereby creating level playing fields and preventing competitive disadvantages for compliant actors (Aslam et al., 2020; Lv et al., 2020). Fourth, standardization systems ensure quality and accountability through technical standards specifying acceptable properties for recycled materials, certification systems that validate capabilities and practices of waste management facilities, and mandatory audit requirements that provide transparency and enable performance monitoring (Gálvez-Martos et al., 2018; Shajidha & Mortula, 2025). Empirical evidence consistently demonstrated that integrated policy approaches combining these multiple instruments proved substantially more effective than single-instrument policies, with jurisdictions employing comprehensive policy packages that achieved construction waste recycling rates of 80-90% compared with rates below 30% in areas with weak or nil policy frameworks (Kabirifar et al., 2020; Shajidha & Mortula, 2025; Wang et al., 2021). International comparisons revealed that countries like Germany, the Netherlands, and Sweden, which implemented strong integrated policies with rigorous enforcement mechanisms, achieved dramatically superior waste management outcomes compared with countries relying primarily on voluntary approaches or weakly enforced regulations (Gálvez-Martos et al., 2018). These findings underscored that the effectiveness of policy depended not merely on existence but on comprehensiveness, integration, and strength of enforcement.

The effectiveness of C&DW management systems should be evaluated holistically through triple bottom line dimensions, covering economic, social, and environmental areas, which collectively capture performance of sustainability and stakeholders' value creation. The economic dimension encompasses multiple-value streams including savings of direct costs from reduced disposal fees. As waste diversion from expensive landfills increases, costs of material procurement decrease through reuse of recovered materials and substitution of recycled materials for virgin resources, enhanced project profitability through improved resource efficiency, and revenue generation from selling recyclable materials such as metals, concrete aggregates, and timber. Empirical studies demonstrated that all-inclusive waste management could reduce overall project costs by 20–40%, while recycling one ton of construction waste saved approximately 37 kg of CO₂ equivalent in emissions and substantial expenses on the procurement of resources (Shajidha & Mortula, 2025). The social dimension reflects impacts on human welfare including creation of employment in the growing recycling and waste processing industries that provide jobs for both skilled and semi-skilled workers. Besides, occupational health and safety could be improved through advanced waste handling practices that reduce risk of injury and exposure to hazardous materials. Community well-being could be enhanced by reducing environmental nuisances such as dust, noise, and traffic from waste transport, as well as increasing stakeholders' awareness of sustainability that creates cultural shifts toward environmental responsibility and thinking of circular economy (Sagan & Mach, 2024; Tafesse et al., 2022; Tan et al., 2024). The environmental dimension captures ecological benefits including significant reduction in greenhouse gas emissions through avoided material production and reduced landfilling decomposition. Landfill burden was decreased by extending facility lifespans and reducing pressures of land use. Natural resources were conserved by

substituting recycled materials for extraction of virgin resources, and broader ecosystem was protected through reduced habitat destruction, water pollution, and soil contamination associated with both resource extraction and waste disposal (Javed et al., 2025; Lee et al., 2024; Mah et al., 2018). The synergistic achievement of superior performance across all three dimensions characterizes truly successful and sustainable implementation of C&DW management. Research indicated that integrated approaches combining technological innovation, supportive policy frameworks, and active stakeholder collaboration yielded optimal outcomes that maximized economic efficiency while enhancing social equity and environmental integrity (Idir et al., 2025; Islam et al., 2024; Purchase et al., 2021). Moreover, the interdependencies among dimensions mean that improvements in one area often catalyze benefits in others. For instance, environmental benefits enhance social legitimacy which reduces regulatory risks and improves economic performance.

The conceptualization of policy as a mediating variable in construction waste management systems represents a critically theoretical and empirical advancement in understanding how improvement initiatives translate into sustainable systemic outcomes. Seminal empirical research by Mohammed et al. (2022) provided compelling evidence that policy factors significantly mediated the relationship between efforts of construction waste management improvement, such as preventive planning, design optimization, and technological innovation and sustainable waste minimization outcomes, while notably demonstrating that policies did not mediate relationships between existing waste generation practices and management outcomes. This crucial distinction reveals that policy's mediating effectiveness is specifically activated by change and innovation rather than by maintenance of status quo practices. It is suggested that policies amplify the impacts of proactive improvement initiatives rather than merely regulating baseline behaviors. The mediating mechanism operates through several interconnected pathways: Policies strengthen and amplify the impacts of preventive planning and innovative practices on waste reduction by providing legitimacy, resources, and enforcement backing (Islam et al., 2024; Mohammed et al., 2022). They also translate successes of localized improvement into scalable regulatory requirements and technical standards applicable across the entire construction sector (Ding et al., 2023; Wang et al., 2021). They provide economic incentives including subsidies, tax benefits, and preferential contracting that accelerate adoption of best practices beyond early adopters to mainstream industry actors (Cheng et al., 2022; Gálvez-Martos et al., 2018). They facilitate stakeholders' collaboration through institutional coordination platforms, information sharing mechanisms, and public-private partnership frameworks that overcome fragmentation and enable collective action (Walters et al., 2024; Zhao, 2021). In developing urban contexts such as Malang, Indonesia, policy's mediating role becomes particularly critical given pervasive challenges like institutional fragmentation across multiple governmental agencies with overlapping mandates. Other barriers include inadequate infrastructure for waste collection, sorting, and recycling, minimal community participation due to low awareness and engagement, and weak coordination between public and private actors (Anisykurlillah et al., 2024; Gunawan, 2025). Research specifically examining Malang emphasized that policy effectiveness as a mediator depended fundamentally on strong enforcement mechanisms that ensured compliance rather than allowing regulations to become paper tigers. Awareness of stakeholders was enhanced and commitment was built through education and participatory processes; infrastructural support was provided for implementing waste management systems including facilities and equipment. Local socio-economic contexts were aligned in recognition of the roles of informal sector and resource constraints (Kurniawan et al., 2022; Unegbu & Yawas, 2024). The transition from traditional command-and-control policy models to collaborative co-creation approaches, coupled with digitalization of waste tracking and monitoring systems, has been identified as essential for enhancing policy effectiveness in mediating between improvement efforts and sustainable C&DW management outcomes, especially in urban contexts like Malang (Gunawan, 2025; Kurniawan et al., 2022). Knowledge and networks of actors in the informal sector are also explicitly recognized and integrated. This body of literature collectively established that understanding policy as a mediating variable rather than merely as an independent influence provided crucial insights for designing interventions. The interference could effectively scale improvement initiatives from project-specific successes to sector-wide transformations toward principles of circular economy in construction.

2.1 Conceptual Framework

Based on the literature review, this study proposed a conceptual framework that positioned policy as a mediating variable between improvement in construction waste management and effectiveness of C&DW management in Malang City. The framework was structured around three main constructs, each measured through specific indicators derived from established theoretical foundations and empirical evidence.

Construction Waste Management Improvement (Independent Variable) represented systematic efforts to enhance waste management practices throughout the project lifecycle. This variable was operationalized through four key phases: (1) Planning—encompassing preventive planning strategies, waste reduction target setting, and material optimization planning (Ajayi et al., 2017; Kabirifar et al., 2020); (2) Design—including waste minimization design, material selection optimization, and design modification to reduce waste (Liu et al., 2020; Shajidha & Mortula, 2025); (3) Construction—involving on-site waste sorting, efficient construction practices,

and real-time waste monitoring systems (Ting et al., 2022); and (4) Operation—focusing on maintenance practices for extended material lifespan and end-of-life material recovery planning (Schützenhofer et al., 2022). These indicators collectively capture the integrated nature of construction waste management improvement across project stages.

Policy (Mediating Variable) served as the regulatory and incentive framework that translated improvement efforts into sustainable outcomes. This construct was measured through four dimensions: (1) Reward and Sanction—performance-based reward systems and penalty mechanisms for non-compliance (Ding et al., 2023; Wang et al., 2021); (2) Incentive—financial incentives including tax reductions, subsidies for recycling facilities, and treatment of preferential procurement (Cheng et al., 2022; Gálvez-Martos et al., 2018; Hua et al., 2022); (3) Regulation—mandatory requirements including waste management plans, sorting obligations, and landfill restrictions (Aslam et al., 2020; Lv et al., 2020); and (4) Standard—technical standards for recycled materials, certification systems, and mandatory audit requirements (Gálvez-Martos et al., 2018; Shajidha & Mortula, 2025). These policy instruments represented the mechanisms through which governmental and institutional actors shaped waste management behaviors and outcomes.

C&DW Management (Dependent Variable) reflected the triple bottom line effectiveness of waste management systems. This variable was conceptualized through three sustainability dimensions: (1) Economic—encompassing cost savings from reduced disposal fees, decreased material procurement costs, and revenue from recyclables (Alsheyab, 2021; Shajidha & Mortula, 2025); (2) Social—including creation of employment in recycling industries, improved occupational health and safety, enhanced community well-being, and increased stakeholders' awareness (Sagan & Mach, 2024; Tafesse et al., 2022; Tan et al., 2024); and (3) Environmental—covering reductions of greenhouse gas emissions, decreased landfill burden, conservation of natural resources, and protection of the ecosystem (Javed et al., 2025; Lee et al., 2024; Mah et al., 2018). The integration of these three dimensions provided a holistic assessment of the effectiveness of C&DW management.

The conceptual framework proposed both direct and indirect pathways. Construction waste management improvement was hypothesized to directly influence policy formulation and implementation, as improved practices created momentum for more effective development of policies (Ding et al., 2023; Wang et al., 2021). Policy, in turn, directly affected C&DW management effectiveness by providing the regulatory structure, economic incentives, and enforcement mechanisms necessary for sustainable outcomes (Islam et al., 2024; Mohammed et al., 2022). In addition, construction waste management improvement was expected to directly impact C&DW management effectiveness through operational efficiencies and implementation of best practices (Javed et al., 2025; Shajidha & Mortula, 2025). Critically, the framework posited that policy served as a mediating variable, channeling and amplifying the effects of improvement efforts on overall waste management effectiveness (Mohammed et al., 2022; Walters et al., 2024). This mediating role is particularly salient in the Malang City context, where challenges of institutional coordination and limited infrastructure demand strong policy frameworks to translate improvement initiatives into measurable outcomes (Anisykurlillah et al., 2024; Gunawan, 2025; Kurniawan et al., 2022).

The proposed conceptual framework illustrating the mediating role of policy between construction waste management improvement and C&DW management performance is presented in Figure 1.

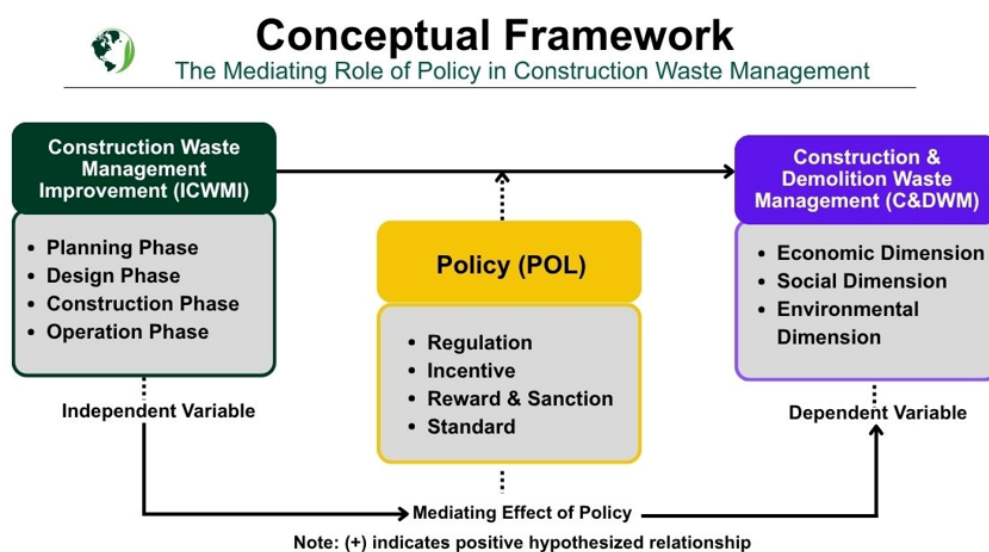


Figure 1. Conceptual framework of the mediating role of policy in construction waste management

Source: Adapted from literature review (Islam et al., 2024; Mohammed et al., 2022; Wang et al., 2021)

2.2 Research Hypotheses

2.2.1 Hypothesis 1: The effect of construction waste management improvement on policy

This hypothesis posited that systematic improvements in construction waste management practices across the planning, design, construction, and operation phases significantly influence the development and strengthening of policy frameworks. Theoretical foundation for this hypothesis emerged from institutional theory and literature of policy innovation, which suggested that improved practices created demonstration effects and generated stakeholders' demand for supportive policy environments (Ding et al., 2023; Wang et al., 2021). When construction projects successfully implement waste reduction strategies, use of efficient materials, and recycling programs, they provide evidence of feasibility and economic viability that policymakers can leverage to justify more stringent regulations and supportive incentives (Gálvez-Martos et al., 2018; Shajidha & Mortula, 2025). Empirical evidence from various contexts demonstrated that jurisdictions with advanced waste management practices tended to develop comprehensive policy packages, including reward and sanction mechanisms, financial incentives, stricter regulations, and higher technical standards (Kabirifar et al., 2020; Wang et al., 2021). In the Malang context, where waste management infrastructure and practices are still developing, improvement initiatives can catalyze policy attention and resource allocation, particularly when they demonstrate tangible benefits in cost savings, environmental protection, and social welfare (Anisykurlillah et al., 2024; Gunawan, 2025). Furthermore, successful improvement practices enhance stakeholders' awareness and advocacy capacity, thus creating bottom-up pressure for policy reform and strengthening (Mohammed et al., 2022). Therefore, this hypothesis proposed that as construction waste management improvement intensified across its four operational phases, policy frameworks would become comprehensive and more enforceable and supportive of sustainable waste management objectives.

H1: Construction waste management improvement has a significantly positive effect on policy.

2.2.2 Hypothesis 2: The effect of policy on construction and demolition waste management

This hypothesis asserted that policy instruments significantly enhanced the effectiveness of C&DW management across economic, social, and environmental dimensions. The theoretical basis derived from regulatory theory and behavioral economics, which emphasized that policies shaped organizational and individual behaviors through combinations of mandates, incentives, and normative frameworks (Cheng et al., 2022; Gálvez-Martos et al., 2018). Policy instruments operating through reward and sanction systems create accountability structures that motivate compliance with waste management requirements, while financial incentives reduce economic barriers to adopting sustainable practices such as recycling and material recovery (Ding et al., 2023; Hua et al., 2022). Regulatory frameworks establish mandatory requirements that standardize waste management practices across projects, to ensure minimum performance thresholds and create level playing fields for industry actors (Aslam et al., 2020; Lv et al., 2020). Technical standards and certification systems provide clear quality benchmarks for recycled materials and waste management processes, to enhance market confidence and facilitate circular material flows (Shajidha & Mortula, 2025). Empirical evidence consistently demonstrated that countries and regions with strong and well-enforced policy frameworks achieved significantly higher recycling rates, lower environmental impacts, and cost-effective economic outcomes compared with those with weak or absent policies (Kabirifar et al., 2020; Shajidha & Mortula, 2025; Wang et al., 2021). In Malang City, where challenges of institutional coordination and limited infrastructure constrain effectiveness of waste management, strengthened policy frameworks encompassing integrated reward-sanction systems, targeted incentives, clear regulations, and enforceable standards are expected to significantly improve C&DW management performance across all three sustainability dimensions (Anisykurlillah et al., 2024; Gunawan, 2025; Kurniawan et al., 2022). This hypothesis therefore predicted that thoroughly and effectively implemented policies would lead to superior economic efficiency, enhanced social benefits, and reduced environmental degradation in construction and demolition-waste management.

H2: Policy has a significantly positive effect on C&DW management.

2.2.3 Hypothesis 3: The effect of construction waste management improvement on construction and demolition waste management

This hypothesis proposed that systematic improvements in waste management practices across project lifecycle phases directly enhanced the overall effectiveness of C&DW management in economic, social, and environmental terms. The theoretical foundation rested on the theory of operational efficiency and resource-based view, which emphasized that process improvements and capability development directly translated into enhanced organizational and system-level performance (Javed et al., 2025). When construction projects implement preventive planning during the planning phase, they reduce waste generation at source, leading to cost savings and resource efficiency (Ajayi et al., 2017; Kabirifar et al., 2020). Design optimization minimizes material waste through explicit specifications and modular approaches, hence directly reducing procurement costs and environmental impacts (Liu et al., 2020). During construction, on-site sorting and efficient practices facilitate

material recovery and recycling, to create economic value while reducing landfill burden and associated environmental harm (Ting et al., 2022; Zhuang et al., 2020). Improvements in the operational phase, including proper maintenance and end-of-life planning, extend material lifespans and enable valuable resource recovery (Schützenhofer et al., 2022). Empirical studies demonstrated that projects implementing comprehensive waste management improvements achieved substantial benefits including 20–40% cost reductions, 50–80% waste diversion from landfills, significant reductions in greenhouse gas emissions, and enhanced stakeholder satisfaction (Alsheyab, 2021; Purchase et al., 2021). In Malang's context, where waste management capacity is still developing, direct improvements in construction practices could yield immediate benefits even in advance of comprehensive policy frameworks. This could address economic pressures on construction firms, create employment opportunities in recycling sectors, and reduce environmental pressures on limited landfill capacity (Kurniawan et al., 2022). This hypothesis therefore predicted that as construction waste management improvement intensified across planning, design, construction, and operation phases, C&DW management effectiveness would improve significantly across economic, social, and environmental dimensions, independent of policy mediation.

H3: Construction waste management improvement has a significantly positive effect on C&DW management.

2.2.4 Hypothesis 4: The mediating role of policy in the relationship between construction waste management improvement and construction and demolition waste management

This hypothesis posited that while construction waste management improvement directly influenced C&DW management effectiveness (as proposed in H3), a significant portion of this effect operated indirectly through policy mechanisms. It is important to note that H3 and H4 examined the same ICWM → C&DW relationship from complementary angles: H3 tested whether a direct path existed, while H4 tested whether that effect was transmitted through policy. Together, they constituted the decomposition of standard mediation model (Hair Jr et al., 2021); a non-significant direct effect (H3) combined with a significant indirect effect (H4) would indicate full mediation—meaning that improvement matters, but only when backed by policy. The theoretical foundation emerged from mediation theory and institutional intermediation frameworks, which suggested that organizational improvements often required institutional translation to achieve systemic impacts (Mohammed et al., 2022; Walters et al., 2024). Policy serves as a critical mediating mechanism by: (1) translating localized improvement successes into scalable regulatory requirements and standards applicable across the construction sector; (2) providing economic incentives that accelerate adoption of improvement practices beyond early adopters; (3) establishing enforcement mechanisms that ensure sustained compliance rather than voluntary or sporadic implementation; and (4) creating institutional coordination platforms that facilitate knowledge transfer and collaborative implementation (Ding et al., 2023; Wang et al., 2021; Zhao, 2021). Empirical evidence specifically demonstrated that policy factors significantly mediated the relationship between improvement factors and sustainable waste minimization outcomes, while notably not mediating relationships between existing practices and outcomes. Therefore, the mediating role of policy is specifically activated by change and innovation rather than maintenance of the status quo (Mohammed et al., 2022). This distinction is particularly relevant in developing urban contexts like Malang, where improvement initiatives face institutional fragmentation, limited infrastructure, and coordination challenges that policy frameworks could help overcome (Gunawan, 2025). The mediating role implies that construction waste management improvements would achieve greater systemic impact when channeled through supportive policy frameworks that provide regulatory backing, economic incentives, technical standards, and enforcement mechanisms (Islam et al., 2024; Kabirifar et al., 2020). Specifically, policies transform improvement initiatives from project-specific successes into sector-wide practices by creating enabling environments, reducing adoption barriers, and ensuring accountability (Gálvez-Martos et al., 2018). This hypothesis therefore predicted partial mediation, where policy significantly channeled and amplified the effects of construction waste management improvement on C&DW management effectiveness, to enhance both the magnitude and sustainability of improvement impacts across economic, social, and environmental dimensions in Malang City.

H4: Policy significantly mediates the relationship between construction waste management improvement and C&DW management.

3. Methods

This study employed a quantitative approach using Partial Least Squares Structural Equation Modeling (PLS-SEM) to examine the mediating role of policy in the relationship between construction waste management improvement and C&DW management in Malang City. The research population comprised construction companies operating in Malang City and surrounding areas. A purposive sampling strategy was employed, selecting 48 companies based on the following criteria: (a) holding an active construction business license (IUJK/SBU); (b) having completed at least one project within the previous two years; and (c) having a designated person responsible for waste-related decisions. One respondent per company was selected, to prioritize the individual most directly involved in waste management decision making. The resulting sample comprised directors ($n = 12$, 25%), project managers ($n = 18$, 37.5%), site engineers ($n = 14$, 29.2%), and project coordinators ($n = 4$,

8.3%). While the sample size of 48 was modest, it met the minimum requirement for PLS-SEM, which was ten times the maximum number of structural paths directed at any single construct (Hair Jr et al., 2021); here the maximum was two paths pointing to C&DW, thus yielding a minimum of 20. Data was collected through structured questionnaires distributed to key personnel who had direct involvement in waste management decision-making. The measurement model consisted of three main constructs: Construction Waste Management Improvement (ICWM) measured through four dimensions (planning, design, construction, and operation), Policy (POL) assessed through four indicators (regulation, incentive, reward and sanction, and standard), and C&DW Management evaluated using three dimensions (economic, environmental, and social aspects). All constructs were measured using reflective indicators on a five-point Likert scale (1 = “Strongly Disagree”, 2 = “Disagree”, 3 = “Neutral”, 4 = “Agree”, 5 = “Strongly Agree”). Questionnaire items were adapted from validated instruments in the literature of construction waste management (Kabirifar et al., 2020; Mohammed et al., 2022; Wang et al., 2021) and were pilot-tested with five industry practitioners for clarity and relevance before full deployment. The wording of the complete item and source attributions are provided in Appendix A.

Evaluation of the measurement model was conducted following the guidelines established by Hair Jr et al. (2021), which emphasized the assessment of reliability and validity through multiple criteria. Convergent validity was assessed using outer loadings (threshold > 0.7), Average Variance Extracted (AVE > 0.5), and composite reliability (>0.7), following the recommendations of Cheah et al. (2018) for reflectively measured constructs. Discriminant validity was evaluated using three approaches: Cross-loadings analysis, Fornell-Larcker criterion, and the Heterotrait-Monotrait (HTMT) ratio with a conservative threshold of 0.85. Reliability of internal consistency was assessed through Cronbach’s alpha and composite reliability coefficients. The structural model evaluation examined path coefficients, statistical significance (*p*-values), coefficient of determination (*R*²), predictive relevance (*Q*²), and effect sizes (*f*²) to determine the strength and significance of hypothesized relationships (Hair Jr et al., 2021).

Because all three constructs were captured through the same questionnaire and the same respondent per company, the model was potentially vulnerable to common method variance (CMV). Several procedural remedies were implemented during data collection: (a) respondent anonymity and confidentiality were guaranteed; (b) the questionnaire separated predictor and criterion items into distinct sections with different response formats and intervening filler questions to reduce consistency motifs; and (c) items were worded to minimize social desirability bias. In terms of statistics, Harman’s single-factor test was conducted by entering all indicators into an unrotated principal component analysis. The first factor accounted for 42.6% of the total variance, below the 50% threshold, thus indicating that CMV did not dominate the data. Besides, the full collinearity variance inflation factor (VIF) values for all constructs were below 3.3 (ICWM = 2.39, POL = 2.85, C&DW = 2.71), which was identified as a conservative threshold for the absence of pathological collinearity attributable to CMV. While these tests could not entirely rule out common method bias, the combined procedural and statistical evidence suggested it was unlikely to substantially distort the reported path estimates.

For hypothesis testing, bootstrapping with 5,000 resamples was performed to assess the significance of path coefficients at the 0.05 level. The effect size (*f*²) was interpreted following Cohen’s guidelines as adopted by Hair Jr et al. (2021): 0.02 (small), 0.15 (medium), and 0.35 (large). The mediation analysis was conducted to test the indirect effect of ICWM on C&DW through policy, with the mediation effect size measured using *ϕ* as recommended by Lachowicz et al. (2018) and further validated by Ogbeibu et al. (2021), with thresholds of 0.01 (small), 0.075 (medium), and 0.175 (large). Model fit was assessed using multiple indices including Standardized Root Mean Square Residual (SRMR < 0.08). Normed Fit Index (NFI; reported for transparency, though Hair Jr et al. (2021) noted that NFI was not a primary fit criterion in PLS-SEM due to its sensitivity to sample size), and other goodness-of-fit measures to evaluate the structural model. Data analysis was performed using SmartPLS 4 software, which was specifically designed for variance-based structural equation modeling and provided all-inclusive tools for both measurement and assessment of the structural model.

4. Results and Discussion

4.1 Descriptive Statistics

The respondents’ profile encompassed various dimensions including gender, age, and educational background. Table 1 presents the descriptive statistics of respondents’ characteristics in this study.

4.2 Measurement Model Evaluation

The measurement model evaluation assessed the validity and reliability of constructs used in this study. Three primary criteria were examined: Outer loadings, which measured the strength of the relationship between indicators and their respective latent variables; Cronbach’s alpha and composite reliability, which assessed internal consistency; and AVE, which evaluated convergent validity. All outer loadings exceeded the threshold of 0.7,

indicating strong indicator reliability. Cronbach's alpha and composite reliability values surpassed 0.9 for all constructs, thus demonstrating excellent internal consistency. AVE values are above 0.8, confirming adequate convergent validity.

Table 1. Descriptive statistics of respondents' characteristics

No.	Characteristics	Category	n	Percent (%)
1	Gender	Male	38	79.17
		Female	10	20.84
Total			48	100.00
2	Age	20–30 years	8	16.67
		31–40 years	21	43.75
		41–50 years	12	25.00
		51–60 years	5	10.42
		>60 years	2	4.17
Total			48	100.00
3	Educational background	High School	4	8.33
		Diploma (D3)	1	2.08
		Bachelor’s Degree (S1)	32	66.67
		Master’s Degree (S2)	10	20.83
		Doctoral Degree (S3)	1	2.08
Total			48	100.00

Note: n = 48 construction companies in Malang City.
Source: Primary data processed, 2025.

Cross-loadings analysis revealed that each indicator loaded most strongly on its intended construct, with values consistently higher than cross-loadings on other constructs. This pattern confirmed discriminant validity, indicating that each construct captured distinct concepts. The Fornell-Larcker criterion further supported discriminant validity, as the square root of AVE for each construct (shown on the diagonal) exceeded its correlations with other constructs. Furthermore, the HTMT ratio values remained below 0.9, satisfying the conservative threshold (Hair Jr et al., 2021; Cheah et al., 2018) and providing additional evidence of discriminant validity. These results collectively demonstrated that the measurement model was robust, reliable, and valid for subsequent structural model analysis.

The results of outer loadings, Cronbach's alpha, composite reliability, and AVE are presented in Table 2.

Table 2. Outer loading, Cronbach's alpha, composite reliability, and Average Variance Extracted (AVE)

Variables and Indicators	Outer Loadings	Cronbach's Alpha	Composite Reliability	AVE
Construction Waste Management Improvement	-			
Construction ← ICWM	0.944	0.966	0.969	0.908
Design ← ICWM	0.950			
Operation ← ICWM	0.954			
Planning ← ICWM	0.963			
Policy	-			
Incentive ← POL	0.947	0.948	0.957	0.865
Regulation ← POL	0.938			
Reward and Sanction ← POL	0.892			
Standard ← POL	0.941			
Construction and Demolition Waste (C&DW) Management	-			
Economic ← C&DW	0.975	0.958	0.959	0.923
Environmental ← C&DW	0.947			
Social ← C&DW	0.960			

Note: All outer loadings > 0.7, Cronbach's alpha > 0.9, Composite reliability > 0.9, and AVE > 0.8, indicating excellent reliability and convergent validity.

Source: Primary data processed using SmartPLS, 2025.

The cross-loadings analysis used to assess discriminant validity is shown in Table 3.

The Fornell-Larcker criterion used to further assess discriminant validity is presented in Table 4.

The HTMT ratio results are reported in Table 5.

Table 3. Cross-loadings analysis

Indicators	Construction and Demolition Waste (C&DW) Management	Construction Waste Management Improvement (ICWM)	Policy (POL)
Economic	0.975	0.712	0.841
Environmental	0.947	0.726	0.870
Social	0.960	0.770	0.807
Construction	0.711	0.944	0.703
Design	0.689	0.950	0.662
Operation	0.756	0.954	0.781
Planning	0.758	0.963	0.745
Regulation	0.869	0.840	0.938
Reward and Sanction	0.735	0.583	0.892
Incentive	0.772	0.641	0.947
Standard	0.857	0.731	0.941

Note: Highlighted values indicate the loading of each indicator on its intended construct. All indicators load highest on their respective constructs, thus confirming discriminant validity.

Source: Primary data processed using SmartPLS, 2025.

Table 4. Discriminant validity—Fornell-Larcker criterion

	Construction and Demolition Waste (C&DW) Management	Construction Waste Management Improvement (ICWM)	Policy (POL)
C&DW Management	0.961	-	-
ICWM	0.766	0.953	-
POL	0.874	0.761	0.930

Note: Diagonal values (highlighted) represent the square root of AVE. Off-diagonal values show correlations between constructs.

Discriminant validity is established as diagonal values exceed all corresponding off-diagonal values.

Source: Primary data processed using SmartPLS, 2025.

Table 5. Heterotrait-Monotrait ratio (HTMT)

	Construction and Demolition Waste (C&DW) Management	Construction Waste Management Improvement (ICWM)	Policy (POL)
C&DW Management	-	-	-
ICWM	0.795	-	-
POL	0.911	0.783	-

Note: The HTMT value for Policy–C&DW was 0.911, which exceeded the strict 0.85 threshold and marginally exceeded the 0.90 threshold (Henseler et al., 2015). While this warranted acknowledgment as a limitation, several lines of evidence suggested that discriminant validity was not critically compromised: (a) the bootstrapped 95% confidence interval for the HTMT (Policy–C&DW) ratio did not include 1.0 (CI = [0.841, 0.967]), satisfying the inferential HTMT criterion (Henseler et al., 2015); (b) the Fornell-Larcker criterion confirmed that the square root of AVE for each construct exceeded all inter-construct correlations (Table 4); and (c) the two constructs were theoretically distinct—Policy captured regulatory/incentive instruments whereas C&DW captured triple bottom line performance outcomes. All remaining HTMT values were well below 0.85. Nevertheless, the proximity of the Policy–C&DW HTMT to the 0.90 threshold should be acknowledged as a limitation, and future research with larger samples should re-examine this boundary.

Source: Primary data processed using SmartPLS, 2025.

4.3 Structural Model Evaluation

The structural model evaluation examined the relationships between constructs and tested the proposed hypotheses. This evaluation included path coefficient analysis, statistical significance testing, effect size assessment (f^2), coefficient of determination (R^2), predictive relevance (Q^2), and overall model fit indices. The path coefficients indicated the strength and direction of relationships between variables, while p -values determined statistical significance. Effect sizes (f^2) measured the substantive impact of predictor variables, and R^2 values assessed the model's explanatory power. Moreover, the Q^2 values evaluated the model's predictive capability, and various fit indices (SRMR, d_{ULS} , d_G , and NFI) to assess the overall adequacy of the structural model.

Figure 2 illustrates the structural model depicting the relationships among Construction Waste Management Improvement (ICWM), Policy (POL), and C&DW Management. The model comprised three main constructs, each measured by multiple indicators represented by yellow rectangles. ICWM was measured through four dimensions: Planning, Design, Construction, and Operation. Policy was assessed through four indicators: Standard, Reward and Sanction, Regulation, and Incentive. C&DW was evaluated using three dimensions: Social, Environmental, and Economic aspects.

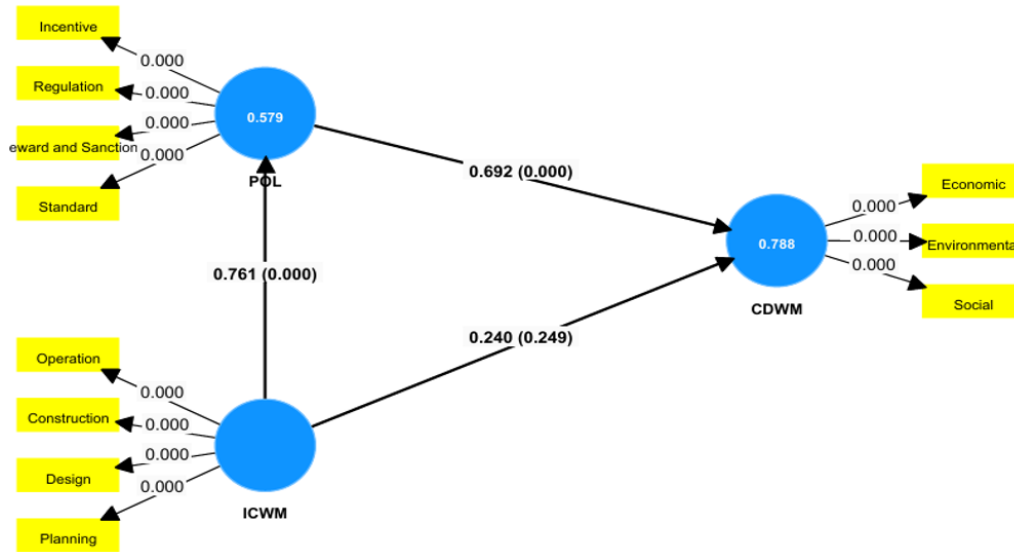


Figure 2. Results of Partial Least Squares Structural Equation Modeling (PLS-SEM) structural model
Source: Primary data processed using SmartPLS, 2025.

The structural relationships were represented by arrows connecting the latent constructs (blue circles). The numbers adjacent to each arrow indicated the path coefficients and their statistical significance (p -values in parentheses). Three direct relationships were hypothesized: ICWM to POL ($\beta = 0.761$, $p < .001$), POL to CDWM ($\beta = 0.692$, $p < .001$), and ICWM to CDWM ($\beta = 0.240$, $p = .249$). The R^2 values displayed within the circles indicated the proportion of variance explained for endogenous constructs. Policy explained 57.9% of its variance ($R^2 = 0.579$), while C&DW demonstrated substantial explanatory power with 78.8% of its variance explained ($R^2 = 0.788$). The indicator loadings displayed as 0.001 in the diagram are a display/export artifact of SmartPLS; actual outer loading values range from 0.892 to 0.975 as reported in Table 2. This structural model configuration allows the testing of both direct effects and the mediating role of policy in the relationship between construction waste management improvement and overall waste management outcomes.

4.3.1 Hypothesis testing

The direct effects among the latent constructs and the results of hypothesis testing are summarized in Table 6.

Table 6. Direct effects

Hypothesis	Path Coefficient	p -Value	95% Confidence Interval		f^2	Result
			Lower Bound	Upper Bound		
H1: ICWM $\rightarrow$ POL	0.761	.001	0.614	0.898	1.374	Supported
H2: POL $\rightarrow$ C&DW	0.692	.001	0.365	1.007	0.950	Supported
H3: ICWM $\rightarrow$ C&DW	0.240	.249	-0.162	0.615	0.114	Not Supported

Note: p -values reported as .001 reflect SmartPLS default rounding to three decimal places and correspond to $p < .001$.

Source: Primary data processed, 2025.

H1 (ICWM $\rightarrow$ POL): The hypothesis was strongly supported ($\beta = 0.761$, $p < .001$). Construction waste management improvement had a strongly positive and significant effect on policy implementation. The f^2 value of 1.374 indicated a large effect size, meaning that ICWM was a crucial predictor of policy adoption. The 95% confidence interval [0.614, 0.898] did not include zero, further confirming the robustness of this relationship. This suggests that improvements in construction waste management practices substantially drive policy formulation and implementation.

H2 (POL $\rightarrow$ C&DW): This hypothesis was also strongly supported ($\beta = 0.692$, $p < .001$). Policy significantly influenced C&DW management outcomes. With an f^2 value of 0.950, the effect size was large, indicating that policy played a critical role in determining waste management effectiveness. The confidence interval [0.365, 1.007] excluded zero, demonstrating statistical robustness. This finding underscores the importance of policy frameworks in shaping sustainable waste management practices.

H3 (ICWM $\rightarrow$ C&DW): This hypothesis was not supported ($\beta = 0.240$, $p = .249$). The direct effect of construction waste management improvement on C&DW management was not statistically significant. The f^2 value of 0.114 indicated a small to negligible effect size. Notably, the confidence interval [-0.162, 0.615] included

zero, suggesting uncertainty in this relationship. This result implies that ICWM may not directly influence C&DW outcomes without the mediating role of policy.

The mediation effect of policy in the relationship between construction waste management improvement and C&DW management performance is presented in Table 7.

Table 7. Mediation effect

Hypothesis	Path Coefficient	p-Value	95% Confidence Interval		Upsilon (ν)	Result
			Lower Bound	Upper Bound		
H4: ICWM → POL → C&DW	0.526	.001	0.299	0.797	0.07654	Supported (Full Mediation)

Note: Upsilon (ν) effect size interpretation based on Cohen via Ogbeibu et al. (2021): 0.01 (Small), 0.075 (Medium), 0.175 (Large). Significance level: $p < .05$.

Source: Primary data processed using SmartPLS, 2025.

H4 (ICWM → POL → C&DW): The mediation hypothesis was strongly supported ($\beta = 0.526$, $p < .001$). Policy serves as a significant mediator in the relationship between construction waste management improvement and C&DW management outcomes. The indirect effect was substantial and statistically significant, with a confidence interval (0.299, 0.797) that excluded zero.

The upslon (ν) value of 0.077 indicated a medium effect size for the mediation, closely approaching the threshold for strong mediation effects. Given that the direct effect (H3) was not significant while the indirect effect through policy was highly significant, this represented full mediation. This meant that ICWM did not directly impact C&DW; instead, its influence was entirely channeled through policy mechanisms. To further substantiate full mediation, the total effect of ICWM on C&DW (direct + indirect) was 0.766 (95% CI [0.645, 0.880], $p < .001$). The variance accounted for Variant Allele Frequency (VAF) was $0.526 / 0.766 = 68.7\%$, which exceeded the 20% threshold commonly cited for partial mediation (Hair Jr et al., 2021). However, VAF alone should not be treated as definitive evidence of full mediation, as it is sensitive to the magnitudes of both direct and indirect effects and can be misleading in small samples (Iacobucci, 2012). Bootstrapped 95% CIs provided more robust evidence: Indirect effect = [-0.162, 0.615] (includes zero, confirming non-significance of the direct path), total effect = [0.645, 0.880] (excludes zero). The primary basis for concluding full mediation is therefore the pattern of a significant indirect effect ($p < .001$), coupled with a non-significant direct effect ($p = .249$), with the VAF serving as supplementary rather than standalone evidence. This finding highlights the critical intermediary role of policy frameworks; improvements in construction waste management practices should be translated into formal policies to effectively enhance overall waste management outcomes.

4.3.2 R^2 and Q^2 analysis

The explanatory and predictive power of the structural model is summarized in Table 8.

Table 8. Explanatory and predictive power of the model

Construct	R^2	R^2 Interpretation	Q^2	Q^2 Interpretation
C&DW	0.788	Substantial	0.711	Strongly Predictive
POL	0.579	Moderate	0.480	Moderately Predictive

Note: R^2 interpretation: 0.25 (Weak), 0.50 (Moderate), 0.75 (Substantial). $Q^2 > 0$ indicates predictive relevance; values > 0.5 suggests strong predictive power.

Source: Primary data processed using SmartPLS, 2025.

4.3.3 Model fit indices

The model fit assessment results are presented in Table 9.

Table 9. Model fit assessment

Fit Index	Saturated Model	Estimated Model	Interpretation
Standardized Root Mean Square Residual (SRMR)	0.061	0.061	Good fit
d_ULS	0.248	0.248	Model stability
d_G	0.611	0.611	Model stability
Chi-square	150.949	150.949	Informative only
Normed Fit Index (NFI)	0.818	0.818	

Source: Primary data processed using SmartPLS, 2025.

5. Discussion

5.1 Construction Waste Management Improvement and Policy Development

The findings of this study provided robust empirical evidence for the critical mediating role of policy in translating construction waste management improvements into effective C&DW management outcomes. The strongly positive relationship between construction waste management improvement and policy formulation (H1: $\beta = 0.761$, $p < .001$, $f^2 = 1.374$) supported propositions of institutional theory that improved practices to generate demonstration effects and stakeholders' demand for supportive policy environments (Ding et al., 2023; Wang et al., 2021). This large effect size indicates that systematic improvements across planning, design, construction, and operation phases serve as catalysts for policy development in Malang City, where waste management infrastructure remains nascent.

Successful implementation of waste reduction strategies, efficient material use, and recycling programs provides policymakers with tangible evidence of feasibility and economic viability, thereby justifying more stringent regulations and supportive incentives (Gálvez-Martos et al., 2018; Shajidha & Mortula, 2025). The mechanism operates through multiple pathways: Best practices become references for regulatory adoption, performance data informs policy evaluation and revision, stakeholders' engagement amplifies advocacy for policy reform, and local successes are replicated as national policy models (Ramos et al., 2023; Wu et al., 2020). This finding contradicts simplistic top-down policy diffusion models, thus revealing a dynamic and reciprocal relationship where ground-level innovations drive institutional change.

5.2 Policy Effectiveness in C&DW Management

The substantial effect of policy on C&DW management (H2: $\beta = 0.692$, $p < .001$, $f^2 = 0.950$) corroborated regulatory theory and behavioral economics frameworks, which emphasized that policy instruments shaped organizational behaviors through mandates, incentives, and normative frameworks (Cheng et al., 2022; Gálvez-Martos et al., 2018; Zhao et al., 2024). This large effect confirmed that comprehensive policy frameworks integrating regulations, incentives, reward-sanction mechanisms, and technical standards significantly enhanced C&DW management across economic, social, and environmental dimensions. The mechanism operates through multiple channels: Reward-sanction systems create accountability structures motivating compliance, financial incentives reduce economic barriers to sustainable practices, regulatory frameworks establish mandatory performance thresholds, and technical standards enhance market confidence in recycled materials (Howell & Moore, 2025; Hua et al., 2022). In Malang's context of institutional coordination challenges and limited infrastructure, these policy instruments appear particularly crucial for overcoming structural barriers (Anisykurlillah et al., 2024; Gunawan, 2025). The finding aligns with comparative international evidence showing that jurisdictions with strong and strictly enforced policies could achieve significantly higher recycling rates and targeted sustainability outcomes (Kabirifar et al., 2020; Wang et al., 2021). Notably, the NFI value of 0.818 fell below the conventional 0.90 threshold. However, Hair Jr et al. (2021) cautioned that NFI was not a reliable fit criterion in PLS-SEM because it (a) penalized model parsimony; and (b) was highly sensitive to sample size, hence producing downwardly biased values for small samples such as $n = 48$. The primary model fit index recommended for PLS-SEM was SRMR, which was 0.061, well within the acceptable 0.08 threshold (Henseler et al., 2015). NFI was reported for transparency rather than as strong evidence of fit.

5.3 Non-Significant Direct Effects and Their Implications

The non-significant direct effect of construction waste management improvement on C&DW management (H3: $\beta = 0.240$, $p = .249$, $f^2 = 0.114$) represented the study's most theoretically significant finding, thus challenging prevailing assumptions about direct operational improvements translating into systemic outcomes. This result suggested that in developing urban contexts characterized by institutional fragmentation and infrastructural constraints, improvement initiatives alone were insufficient to drive sustainable waste management transformation (Anisykurlillah et al., 2024; McCormick et al., 2013). While theory of operational efficiency and resource-based view predict that process improvements directly enhance performance (Javed et al., 2025), the Malang case reveals that localized successes remain project-specific without institutional translation mechanisms. This finding resonates with Mohammed et al. (2022), who demonstrated that policy factors mediated relationships between improvement initiatives but not between existing practices and outcomes. This indicates that policy's mediating role activates specifically during change processes rather than maintaining status quo. The absence of direct effects implies that benefits from planning optimization, design efficiency, construction-phase sorting, and operational improvements dissipate at project boundaries without policy frameworks to scale and sustain them across the sector.

5.4 Policy as a Full Mediator

The full mediation effect ($H4: \beta = 0.526, p < .001, v = 0.077$) provided critical insights into policy's institutional intermediation role in waste management systems. The medium effect size indicates that policy serves as more than a passive conduit, actively amplifying and channeling improvement effects through four mechanisms: (1) translating localized successes into scalable regulatory requirements applicable sector-wide; (2) providing economic incentives accelerating adoption beyond early adopters; (3) establishing enforcement mechanisms to ensure sustained rather than sporadic compliance; and (4) creating coordination platforms to facilitate knowledge transfer and collaborative implementation (Wang et al., 2021; Zhao, 2021). The full mediation structure, where indirect effects are significant while direct effects are not, demonstrates that improvement initiatives achieve systemic impact only when embedded within supportive policy frameworks (Islam et al., 2024). Several Malang-specific factors help explain why the direct path from improvement to outcomes is entirely absorbed by policy. First, C&DW regulatory framework in Malang remains fragmented across multiple agencies (Dinas Lingkungan Hidup, Dinas PUPR, and DPRD committees) with overlapping mandates and no single authority responsible for construction waste streams (Anisykurlillah et al., 2024). Improvements of individual company remain isolated without a coordinating policy mechanism. Second, the informal sector plays a dominant role in Malang's waste recycling ecosystem; waste pickers and small-scale recyclers operate outside formal regulatory oversight. Market-based improvements by construction firms cannot connect with downstream recycling infrastructure unless policy bridges that gap (Kurniawan et al., 2022). Third, inter-firm trust is low in Malang's competitive construction market, where firms are reluctant to share best practices without regulatory assurance, thus constituting a typical problem of collective action only policy can resolve (Gunawan, 2025). Fourth, enforcement of existing regulations (e.g., AMDAL requirements) is widely perceived as weak, meaning efforts of voluntary improvement offer no competitive advantage unless policy enforcement levels the playing field. These factors explain why improvement efforts require policy "activation" to generate systemic outcomes. This finding has profound implications for developing cities pursuing circular economy transitions: Investment in operational improvements should be accompanied by parallel policy development to capture and scale benefits. The model's substantial explanatory power ($R^2 = 0.788$ for C&DW, $R^2 = 0.579$ for policy) and strong predictive relevance ($Q^2 = 0.711$, $Q^2 = 0.480$) validate these relationships, while the SRMR value of 0.061 confirms good model fit.

6. Conclusions

This study established that policy served as a critical mediating mechanism through which construction waste management improvement translated into effective C&DW management outcomes in developing urban contexts. The full mediation structure revealed that operational improvements in planning, design, construction, and operation phases significantly influenced policy development, which in turn substantially enhanced waste management effectiveness across economic, environmental, and social dimensions. However, in view of the absence of policy frameworks, these improvements remained confined to individual projects without achieving systemic transformation. The findings demonstrated that 78.8% of variance in C&DW management effectiveness could be explained by the combined pathways of improvement-to-policy and policy-to-outcomes, with policy accounting entirely for the relationship between improvement initiatives and results of sustainable waste management. These results underscored that in contexts characterized by institutional fragmentation and infrastructural constraints such as Malang City, integrated policy instruments encompassing regulations, incentives, reward-sanction mechanisms, and technical standards were indispensable for scaling localized successes into sector-wide sustainable practices. The research contributes to mediation theory and institutional intermediation frameworks by empirically demonstrating that the role of policy extends beyond passive regulation to active amplification of improvement effects, thereby bridging micro-level innovations and macro-level systemic change in waste management systems.

7. Implications

7.1 Theoretical Implications

The theoretical implications of this research advance understanding of policy mediation in sustainability transitions by demonstrating that institutional frameworks do not merely facilitate but fundamentally transform how operational improvements generate systemic outcomes. The full mediation structure challenges direct implementation models prevalent in literature of operational efficiency, thus revealing that in fragmented institutional contexts, policy serves as an essential translation mechanism converting project-level innovations into sector-wide practices.

7.2 Practical Implications

Practically, these findings provide actionable guidance for policymakers and stakeholders in the construction industry in developing cities to pursue objectives of the circular economy. Government authorities should prioritize the development of comprehensive and integrated policy packages to combine regulatory mandates, economic incentives, performance standards, and enforcement mechanisms rather than relying solely on voluntary industry initiatives. Construction firms should actively engage in policy advocacy and development processes, recognizing that operational improvements yield maximum returns when embedded within supportive regulatory frameworks. For Malang City specifically, the results suggested prioritizing policy strengthening alongside capability building initiatives, to ensure that emerging best practices in waste reduction, recycling, and material efficiency are institutionalized through formal regulatory channels, financial incentive structures, and technical standard-setting mechanisms to achieve sustainable and scalable waste management transformation.

8. Limitations and Future Research

Future research should extend this framework through longitudinal design to examine how the relationships among improvement, policy, and outcomes evolve over time as institutional capacity matures. It could also potentially reveal threshold effects where direct pathways emerge alongside the strengthening of policy ecosystems. Comparative studies across multiple cities with varying institutional development levels would illuminate boundary conditions for policy mediation, to test whether full mediation persisted universally or transitioned to partial mediation in more developed contexts. Investigating moderating variables such as enforcement capacity, stakeholder coordination mechanisms, availability of infrastructure, and organizational capabilities would provide nuanced understanding of conditions enhancing or constraining the effectiveness of policy's mediating role. Methodologically, mixed-method approaches combining PLS-SEM with qualitative case studies would unravel specific policy mechanisms (regulation vs. incentive vs. standard) driving outcomes, while experimental or quasi-experimental designs could establish causal relationships beyond cross-sectional correlational evidence. Expanding the model to incorporate additional mediators (e.g., technological innovation, stakeholder collaboration, and market mechanisms) and outcomes (e.g., carbon emissions, resource circularity, and economic competitiveness) would provide in-depth understanding of the dynamics of waste management system. Finally, further studies replicating different national and cultural contexts could test its theoretical generalizability and identify context-specific factors in shaping the transitions of policy-mediated sustainability in the construction sector globally.

Author Contributions

Conceptualization, H.S.Y.P. and A.; methodology, H.S.Y.P.; software, H.S.Y.P.; validation, H.S.Y.P., A., and A.E.; formal analysis, H.S.Y.P.; investigation, H.S.Y.P.; resources, A. and W.; data curation, H.S.Y.P.; writing—original draft preparation, H.S.Y.P.; writing—review and editing, A., A.E., and W.; visualization, H.S.Y.P.; supervision, A., A.E., and W.; project administration, H.S.Y.P.; funding acquisition, H.S.Y.P. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflict of interest.

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Appendix

Appendix A. Questionnaire items, and construct mapping

Code	Item Wording
Construct 1: Construction Waste Management Improvement (ICWM)	
<i>Dimension: Planning (PLN)</i>	
PLN1	The company conducts surveys and analyses of existing site conditions to plan waste management strategies.
PLN2	The planning team routinely identifies potential waste sources at project sites before construction begins.
PLN3	The company effectively prepares timelines and schedules for waste management at each construction phase.
PLN4	The company integrates waste management programs into overall project planning.
<i>Dimension: Design (DES)</i>	
DES1	The design team effectively creates designs that minimize construction waste production.
DES2	Design review processes thoroughly evaluate waste management aspects.
DES3	Structural analysis is conducted in depth to optimize material use and reduce waste.
DES4	The company validates designs in accordance with building codes that regulate waste management.
<i>Dimension: Construction (CON)</i>	
CON1	The company effectively uses 3D coordination technology (e.g., BIM) to minimize construction waste.
CON2	Construction site layout planning considers areas for waste management and sorting.
CON3	Construction sequencing is planned to minimize waste generation.
CON4	The company uses modular or prefabricated construction systems to reduce waste.
<i>Dimension: Operation (OPR)</i>	
OPR1	The maintenance scheduling system effectively reduces operational waste.
OPR2	Building system performance analysis optimizes operations and reduces waste.
OPR3	Building asset analysis considers lifecycle optimization and waste reduction.
OPR4	The waste reporting and monitoring system during building operations is accurate and of high quality.
Construct 2: Policy (POL)	
<i>Dimension: Regulation (REG)</i>	
REG1	The company understands and consistently applies PP No. 22/2021 on environmental protection and management in construction waste management.
REG2	Implementation of Minister of PUPR Regulation No. 18/2021 is effective for construction waste management.
REG3	The company complies with local regulations governing construction waste management.
<i>Dimension: Standard (STD)</i>	
STD1	The company consistently applies green building certification standards (e.g., LEED, BREEAM, Greenship).
STD2	The company implements ISO 14001 environmental management system standards.
STD3	The company consistently applies construction industry standards related to waste management.
<i>Dimension: Incentive (INC)</i>	
INC1	The company effectively utilizes available fiscal incentives for environmentally friendly waste management practices.
INC2	The company actively participates in environment-related Corporate Social Responsibility (CSR) programs.
INC3	The company has good access to government-provided technical assistance and training programs for waste management.
<i>Dimension: Reward and Sanction (RWS)</i>	
RWS1	The internal reward system effectively encourages good waste management practices.
RWS2	The company's risk management system effectively anticipates government sanctions related to waste management.
RWS3	The reward and sanction system significantly influences the company's motivation to improve waste

management performance.

Construct 3: Construction and Demolition Waste Management (C&DW)

Dimension: Environmental (ENV)

- | | |
|-------------|--|
| ENV1 | The company effectively reduces greenhouse gas emissions through C&DW management practices. |
| ENV2 | C&DW management effectively reduces smog formation and minimizes air pollution contributions. |
| ENV3 | The company effectively minimizes land use for waste disposal through reduce, reuse, and recycle programs. |
-

Dimension: Economic (ECO)

- | | |
|-------------|---|
| ECO1 | C&DW management efficiently controls material and equipment costs. |
| ECO2 | The waste transportation system is highly cost-efficient. |
| ECO3 | Material reuse practices generate significant cost savings. |
| ECO4 | The recycling program produces substantial cost savings. |
| ECO5 | The value recovery system effectively enhances the economic value of waste materials. |
-

Dimension: Social (SOC)

- | | |
|-------------|---|
| SOC1 | The company effectively protects the health of workers involved in C&DW management. |
| SOC2 | Comprehensive occupational safety protocols are applied in waste management operations. |
| SOC3 | C&DW management minimizes negative health impacts on surrounding communities. |
| SOC4 | An integrated safety system is implemented across all waste management activities. |
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Note: All items were measured on a five-point Likert scale (1 = “Strongly Disagree,” 2 = “Disagree,” 3 = “Neutral,” 4 = “Agree,” 5 = “Strongly Agree”). Items were originally administered in Indonesian (Bahasa Indonesia) and translated into English for reporting purposes. Each dimension score is computed as the mean of its constituent items. Item wording has been adapted from the cited sources to fit the construction waste management context in Malang City, Indonesia.